

DR. SUNEETHA MEKALA, MD, MS

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PROFESSIONAL SUMMARY

Research Data Analyst with an **M.D. (IMG), M.S. in Health Data Science**, and **5+ years** of experience managing, integrating, and analyzing large-scale clinical, epidemiologic, **population health, aging**, and longitudinal research datasets supporting **NIH-funded** and investigator-led studies. Experienced in designing reproducible data management workflows, developing analysis-ready datasets, and performing statistical analyses and programming using **R, Stata, SAS, SQL, and Python**. Skilled in integrating data from **Cerner PowerChart EHR, REDCap, laboratory systems, registries, administrative healthcare databases, and Google BigQuery**, with expertise in ETL, data validation, quality assurance, data governance, and reproducible research. Proven ability to collaborate with investigators, clinicians, epidemiologists, biostatisticians, and multidisciplinary research teams to investigate clinical outcomes, population health, and aging-related research questions, while supporting scientific publications, grant-funded research, conference presentations, and data-driven decision-making.

EXPERIENCE

Dell Medical School, UT Austin

Mar 2025 – Present

Data Analyst - Population Health

Austin, TX

- Develop and maintain secure, analysis-ready research databases by integrating large-scale clinical and population health data from **Cerner PowerChart EHR, REDCap, SQL** databases, registries, administrative systems, and **Google BigQuery** to support **five NIH- and CPRIT-funded cancer prevention and population health research programs**, enabling investigator-led studies and evidence-based public health initiatives.
- Design complex **SQL** queries and ETL workflows to extract, clean, validate, and harmonize data from multiple sources, ensuring data accuracy, completeness, and reproducibility across research projects involving **50,000+ patient records**.
- Perform descriptive, inferential, and longitudinal statistical analyses using **Python, R, SAS, Stata, and SQL**, generating publication-ready tables, figures, and analytical reports that support population health research and evidence-based decision making.
- Develop customized dashboards, reports, and visualizations using **Tableau** and **Power BI** to communicate research findings, study metrics, and operational performance to investigators, clinicians, and multidisciplinary collaborators.
- Implement comprehensive data validation, quality assurance, cross-source reconciliation, and standardized documentation procedures to maintain high-quality research databases, support regulatory compliance, and resolve data quality issues.
- Collaborate with physicians, principal investigators and research teams to define data requirements, support new research initiatives, respond to data requests, and contribute to scientific presentations, grant reporting, and project deliverables.

Regenstrief Institute

Jun 2024 – Feb 2025

Data Analyst - IU Aging Research

Indianapolis, IN

- Managed and analyzed large-scale longitudinal clinical and epidemiologic datasets to support **NIH-funded** aging and cognitive health research, integrating data from **Cerner PowerChart EHR, REDCap**, registries, and administrative healthcare databases.
- Performed descriptive, inferential, and longitudinal statistical analyses using **Stata, SQL, and Python** to investigate aging-related clinical outcomes, healthcare utilization, chronic disease risk factors, and population health trends.
- Developed analysis-ready research datasets through data extraction, harmonization, cleaning, validation, and quality assurance, ensuring reproducible workflows and high data integrity across multiple longitudinal studies.
- Conducted data quality assessments, cross-source reconciliation, missing data evaluation, and comprehensive documentation to support reproducible epidemiologic analyses and scientific research.
- Collaborated with physicians, epidemiologists, biostatisticians, and multidisciplinary investigators to support statistical analyses, scientific manuscripts, conference abstracts, and presentations for aging and population health research.
- Contributed to investigator-initiated studies by providing methodological support, statistical reporting, and data management for observational and longitudinal research projects examining aging and chronic disease.

Indiana University School of Medicine

Mar 2023 – Sep 2024

Research Data Analyst

Indianapolis, IN

- Supported **NIH-funded** research in **gastroenterology and liver diseases** by leading **biostatistical analyses in R, Stata, and SAS** and translating results into publication-quality tables, figures, and interpretive summaries for investigators.
- Built and maintained **analysis frameworks** and reproducible workflows (cleaning, QC checks, variable derivation, analytic plans) to enable consistent reporting across projects and accelerate turnaround for time-sensitive deliverables.
- Developed tailored **cohort datasets** using clinical, financial, and operational criteria, including harmonization of **international datasets** to support comparative analyses; improved disease correlation insights by **30%** and reduced interpretation time by **40%**.

- Partnered with physician-led research teams and hospital stakeholders to define research questions, align timelines, and deliver ad hoc and recurring analytic outputs; contributed to **manuscripts** and **grant submissions** through methods/results drafting and literature synthesis.
- Amrita Hospitals — Clinical Research

Feb 2021 – Nov 2022

Physician - Clinical Research Scientist

India

 - Oversaw study implementation (enrollment, data collection, analysis) and conducted routine data audits to ensure protocol adherence and data consistency.
 - Supported adherence to regulatory/ethical guidelines, including **HIPAA, FDA, and IRB** requirements, protecting participant confidentiality and data integrity.

EDUCATION

- Indiana University

Jan 2023 – Present

Master of Science in Health Informatics (GPA: 3.8)

Indianapolis, IN
- National Pirogov Memorial Medical University

Sep 2011 – Jun 2017

M.D. in Medicine (GPA: 3.7)

Vinnytsia, Ukraine

SKILLS

- **Statistical Programming:** SAS, R, SQL (MySQL, PostgreSQL), Python (Pandas, NumPy, SciPy, Scikit-learn), STATA, SPSS
- **Data Management & Integration:** ETL, Data Extraction, Data Integration, Data Harmonization, Data Cleaning, Data Validation, Data Quality Assurance (QA), Data Pipelines, Longitudinal Data Management, Analysis-Ready Dataset Development
- **Clinical Research & Healthcare Data:** Cerner PowerChart EHR, REDCap, OpenEMR, Laboratory Data, Registry Data, Administrative Healthcare Data, Google BigQuery, ICD-10, SNOMED CT, LOINC, HL7, FHIR, OMOP
- **Statistical Analysis & Reporting:** Descriptive Statistics, Inferential Statistics, Longitudinal Data Analysis, Epidemiologic Analysis, Reproducible Research, Statistical Reporting
- **Visualization & Reporting:** Tableau, Power BI, Excel, ggplot2, Matplotlib, Seaborn
- **Data Governance & Compliance:** HIPAA, Data Governance, Metadata Management, Data Dictionaries, Regulatory Compliance, Good Clinical Practice (GCP)
- **Development Tools:** Git, GitHub, Jupyter Notebook, Google Colab, VS Code
- **Machine Learning:** Machine Learning, Natural Language Processing (NLP), Large Language Models (LLMs)

PROJECTS

- Fracture Outcome Prediction Using ML Models

Feb 2023 – May 2023

 - Built supervised models (**Decision Tree, Random Forest, Logistic Regression**) achieving best **AUC = 0.94**; applied **SMOTE** and feature engineering in **SQL/Python**.
 - Identified key predictors (e.g., BMI, femoral neck thickness) to inform targeted prevention strategies.
- Statistical Assessment of Mortality Risk Among ICU-Admitted Heart Failure Patients

Sep 2023 – Dec 2023

 - Applied **Mann–Whitney U, Chi-square**, and **logistic regression** in **R/STATA** to evaluate factors associated with cardiovascular outcomes.
 - Used **R, STATA, and Tableau** to assess ventilation–mortality associations and communicate findings through clear visualizations and reporting.
- Predicting Fetal Health from Cardiotocogram Data Using ML Models

Jan 2024 – May 2024

 - Implemented preprocessing with **StandardScaler** and **SMOTE** to address class imbalance and improve model performance.
 - Performed feature selection in **Python**, reducing dimensionality and cutting computational time by **30%** while maintaining predictive accuracy.
- Parkinson’s Disease Detection Using Machine Learning

Sep 2024 – Oct 2024

 - Optimized **SVM** models for vocal feature analysis, achieving **87.18% accuracy**, **93.55% recall**, and **90.63% precision**.
 - Validated findings with **Shapiro–Wilk** and **Mann–Whitney U** tests to support robustness and clinical relevance.
- Sober Steps Mobile Health Application

Jan 2023 – May 2023

 - Led end-to-end development of an award-winning mHealth project addressing alcohol use disorder; designed a user-centered app supporting behavior change and recovery.

CERTIFICATIONS

- Power BI for Data Analysts

Sep 2024

LinkedIn
- Responsible Conduct of Research

May 2024

CITI Program

Good Clinical Practice <i>CITI Program</i>	Mar 2024
Biomedical Researcher <i>CITI Program</i>	Feb 2023
SAS 9.4 Cert Prep: SAS Programming Essentials <i>SAS Institute</i>	Apr 2023
Intermediate SQL <i>DataCamp</i>	Jan 2023

AWARDS

- Best Project Certificate**
INFO B505: Informatics Project Management
 - Managed the full project life cycle from research and design through implementation, earning the **Best Project Certificate**.
 - Recognized for project execution, user experience design, and impactful contributions to addiction recovery solutions.

VOLUNTEERING

- American Medical Informatics Association (AMIA)**
Volunteer — AMIA 2024 Annual Symposium

Nov 2024
San Francisco, CA

 - Supported on-site symposium operations and attendee assistance for the AMIA 2024 Annual Symposium.

PUBLICATIONS

- Relationship between Intensive Care Unit Delirium and Change in Quality of Life, Mood, and Cognition Over 12 Months in Acute Respiratory Failure Survivors** — *CHEST Critical Care*. Co-author; performed advanced data and statistical analysis.
- Metabolic Dysfunction Associated Steatotic Liver Disease in Individuals with and without HIV in Kenya, In Review** — Co-author; performed advanced data and statistical analysis - In review
- Elevated Liver Biochemistries during Pregnancy: Frequency, Clinical Characteristics, and Etiologies, In Review** — Co-author; conducted advanced data and statistical analysis - In review
- Association of Diabetes Risk Index with Metabolic Liver Disease and Fibrosis in Individuals with PCOS, In Review** — Co-author; conducted advanced data and statistical analysis - In review